

# Hernán Salinas Castro

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## Interests

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Bachelor's degree graduate in Astronomy with experience in the study of exoplanets using the transit method and sampling with the Markov Chain Monte Carlo (MCMC) algorithm. Interested in learning new techniques for the analysis of exoplanetary atmospheres, with the goal of working on future projects using telescopes such as the ELT and ARIEL.

## Education

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**Pontificia Universidad Católica de Chile**

*PhD in astrophysics*

Santiago, Chile

2026 - Present

**Universidad Andrés Bello**

*Bachelor's degree in astronomy*

Santiago, Chile

2020 - 2024

## Research experience

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**Visitor at ESO (Santiago)**

June 2025 - Present

Study of exoplanetary atmospheres through high-resolution spectroscopy techniques. Conducted calibration and data analysis of observations obtained with the HARPS and NIRPS spectrographs, focusing on a hot Jupiter-type exoplanet in a close-in orbit around its host star. This work is carried out under the supervision of Dr. Elyar Sedaghati.

**VLTI Observations of the Atmosphere of Normal Red Giants**

January 2025 -

June 2025

Used AMBER data from the VLTI to study red giant stars through optical and infrared interferometry techniques. During this project, I had the opportunity to learn how to calibrate and analyze interferometric data under the supervision of Dr. Keiichi Ohnaka.

**Member of the MOTE Project | Monitoring of Transiting Exoplanets (MOTE) Project**

December 2023 – Present

Participated in a project focused on training undergraduate students in the analysis of exoplanet transit data. I contributed to the selection and planning of observation nights for target exoplanets using different telescopes, such as SPECULOOS and LCOGT. I was also involved in the calibration and reduction of the collected data, followed by photometric analysis. As part of this work, I developed a sub-project focused on the study of Jupiter-type exoplanets orbiting M-dwarf stars, which became the topic of my undergraduate thesis. Additionally, I had the opportunity to write an observation proposal during

my undergraduate studies as part of this sub-project. This entire project was carried out under the supervision of Dr. Claudio Cáceres.

## Workshops and courses

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| <b>NASA Exoplanet Science Institute</b>       | Virtual          |
| <i>Silver Jubilee: Exoplanet Demographics</i> | July 21-25, 2025 |

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| <b>European Interferometry Initiative (EII)</b>          | Virtual             |
| <i>Horizons for Optical Long Baseline Interferometry</i> | January 27-30, 2025 |

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| <b>Universidad Andrés Bello</b> | Santiago, Chile         |
| <i>Radio astronomy course</i>   | August - December, 2024 |

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| <b>Universidad Andrés Bello</b>                      | Santiago, Chile         |
| <i>Complementary cosmological simulations course</i> | August - December, 2024 |

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| <b>Space Mood</b>                             | Virtual           |
| <i>Space Biology Networking in Peru (BEP)</i> | December 14, 2024 |

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| <b>Universidad Andrés Bello</b>      | Santiago, Chile  |
| <i>First Astrochemistry Workshop</i> | October 28, 2024 |

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| <b>NASA Exoplanet Science Institute</b>                                    | Virtual          |
| <i>Advances in Direct Imaging: From Young Jupiters to Habitable Earths</i> | July 22-26, 2024 |

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| <b>24RED</b>                                        | Virtual   |
| <i>The virtual astrobiology introductory course</i> | June 2024 |

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| <b>Centro de Astrofísica y Tecnologías Afines (CATA)</b> | Santiago, Chile |
| <i>CATA Area 6 (Exoplanetas y astrobiología) Meeting</i> | May 7, 2024     |

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| <b>Universidad Andrés Bello</b>            | Santiago, Chile   |
| <i>Complementary interferometry course</i> | March - June 2024 |

## Talks

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| <b>Centro de Astrofísica y Tecnologías Afines (CATA)</b> | Santiago, Chile |
| <i>CATA Area 6 Meeting</i>                               | August 6, 2025  |
| <i>"Jupiter type exoplanets orbiting M dwarf stars"</i>  |                 |

## Outreach activities

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### **Member of the Astroamigos Wintata Wara Group**

April 2025 - Present

A astronomy group engaged in public science outreach through activities such as telescope observations, public talks, interactive stations, and workshops. Within the group, I serve as a science communicator and collaborator, supporting content development and assisting with telescope operations. I have participated in events at institutions such as Observatorio Calán, Observatorio Manuel Foster, and Museo Nacional Aeronáutico y del Espacio. Instagram: @astroamigos\_wintata\_wara.

### **Galaxies for Children Workshop**

May 2023

Together with two colleagues, I organized an outreach workshop for children at Luis Cruz Martínez School in Andacollo, Chile. We introduced basic concepts about galaxies and guided an interactive activity where the children created their own galaxies using plasticine. The goal was to foster creativity, curiosity, and interest in astronomy.

## Technical Skills

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**Languages:** Spanish - Intermediate English.

**Programming language:** Python and LaTeX(intermediate level).

Bash script (Basic level).

**Software:**

Intermediate level: Topcat, Aladin, SAOImageDS9.

Basic level: Excel, Iraf, Astroimagej, Casa, Carta, FV.

**TESS telescope data analysis**(Basic level).

**Photometric data analysis for exoplanets:**

Intermediate level: exotic\_ld, ccdproc, batman, lightkurve, emcee.